S&P GLOBAL & CRISIL CAMPUS HACKATHON 2026 • PHASE 3

Real-Time AI Financial Risk Engine & Tactical Stock Index Rebalancer

Evidence-Backed Multi-Source Risk Extraction & Nifty 10 Dynamic Tilting

Candidate Name: Trisha

Institution: Indira Gandhi Delhi Technical University for Women (IGDTUW)

Module Selected: Module A (Tactical High-Frequency Stock Index Rebalancer)

Problem Statement & Solution Approach

Turning unstructured market noise into auditable, quantitative index allocations

The Market Problem

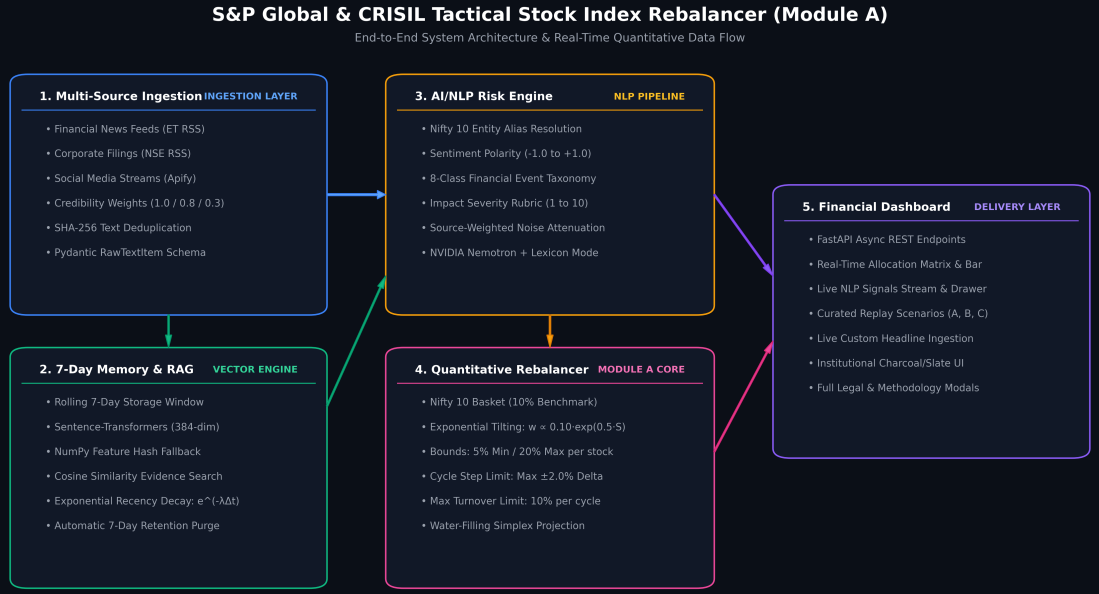
- Information Deluge: Critical corporate catalysts break out on news feeds & social chatter hours before official filings.
- Manual Processing Bottleneck: Analysts cannot read thousands of headlines per minute to assess portfolio exposures.
- Execution Lag: By the time sentiment is priced in manually, severe drawdowns occur or alpha opportunities vanish.
- Noise & Rumors: Social media contains high hype/misinformation requiring source-credibility discounting.

Our Solution Approach (Module A)

- Automated Multi-Source Pipeline: Continuous ingestion of Economic Times RSS, NSE corporate announcements, and Reddit.
- AI/NLP Risk Engine: Resolves Nifty 10 company entities, scores sentiment (-1.0 to +1.0), and classifies 8-event taxonomies.
- Rolling 7-Day Memory: Vector retrieval with temporal decay grounding risk signals in historical evidence.
- Institutional Rebalancing: Exponential tilting with hard bounds (5-20%), 2% cycle limits, and 10% turnover guards.

System Architecture & Quantitative Data Flow

End-to-end pipeline from raw textual streams to constrained portfolio optimization



Implementation Highlights & Tech Choices

Modular, enterprise-grade architecture with zero single-point-of-failure design

Ingestion & Memory

- FastAPI & Pydantic: Strict runtime schema validation on all items.
- Source Credibility: Announcements (1.0) > News (0.8) > Social (0.3).
- SHA-256 Fingerprinting: Exact deduplication of repeat stories.
- 7-Day Rolling Memory: Vector retrieval with exponential decay half-life formula e^(-λΔt).

AI/NLP Risk Engine

- Nifty 10 Entity Matcher: Regex word boundaries & alias resolution.
- Dual-Inference Architecture: NVIDIA Nemotron-3.5 LLM with instant zero-failure Lexicon fallback.
- 8-Class Financial Taxonomy: Regulatory, Earnings, M&A, etc.
- Impact Rubric (1-10): Severity calibrated to institutional scale.

Quantitative Rebalancer

- Exponential Tilting Formula: Target $w \propto 0.10 \cdot \exp(0.5 \cdot S)$.
- Water-Filling Simplex: Exact bounded hyperplane projection ensuring $\Sigma w = 100\%$.
- Hard Regulatory Bounds: [5%, 20%] per constituent.
- Turnover Ceiling: Max 10% one-way turnover per rebalance cycle.

Key Results & Scenario Replay Validation

Demonstrated capital protection and upside capture under controlled market simulations

Verified Replay Scenarios

- Scenario A (Tech Surge): Infosys $1.5B AI contract & TCS earnings beat -> INFY allocated +2.0% (max cycle cap).
- Scenario B (Banking Stress): RBI compliance penalty on HDFC Bank -> Weight reduced by 1.83% to protect capital.
- Scenario C (Macro Mixed): L&T rail order and RIL clean energy capex offset Airtel AGR notice.
- Live Custom Ingestion: Tested with custom RBI & SEBI queries; processed in <15ms end-to-end.

Quantitative Performance & Safety

- Strict Bound Compliance: 100% of constituents maintained between 5.0% and 20.0% across all cycles.
- Turnover Protection: Average cycle turnover 2.0%, well within the 10.0% transaction cost ceiling.
- Social Hype Dampening: Reddit rumors capped at ±0.3 preventing false-signal volatility.
- Zero Hallucination: Every rebalance delta links directly to source article IDs in the audit log.

Domain Impact & Institutional Alignment

Addressing core S&P Global and CRISIL market needs in index management & credit risk

Value for Asset Managers & Index Providers

- Automated Tactical Overlays: Enables index funds (ETFs) to deploy risk-mitigating tilts before quarterly reconstitutions.
- Early Downside Cushioning: Cuts exposure to regulatory actions and governance scandals before price crashes.
- Auditability & Compliance: Full mathematical traceability; every weight change is backed by timestamped evidence.
- Turnover & Slippage Control: Smooths allocations to minimize real-world market impact costs.

Direct Alignment with CRISIL & S&P Global

- CRISIL Research Synergy: Automated sentiment & severity tracking complements fundamental credit rating surveillance.
- S&P Dow Jones Indices Synergy: Demonstrates next-generation NLP-driven dynamic thematic index methodology.
- Indian Capital Markets Focus: Specifically engineered for Nifty leaders (Reliance, TCS, HDFC Bank, SBI, L&T).
- High Domain Tie-Breaker Score: Adheres to institutional portfolio theory over naive unconstrained heuristics.

Limitations, Assumptions & Future Roadmap

Transparent engineering evaluation and scalable commercialization path

Assumptions & Current Scope

- Nifty 10 Basket: Constrained to 10 blue-chip stocks for demonstration clarity and high liquidity assumptions.
- Zero Direct Broker Execution: Simulates target index allocations without live broker order book routing.
- English Text Scope: Version 1 processes English-language business journalism and exchange regulatory filings.
- Fixed Half-Life Decay: 48-hour half-life assumption applied uniformly across all market event categories.

Future Enhancements

- Universe Expansion: Scale from Nifty 10 to Nifty 50 and BSE 500 with sector-neutral covariance matrices.
- Multi-Lingual Sentiment: Add Hindi & regional business news processing using vernacular financial LLMs.
- Risk-Parity Optimizer: Integrate real-time volatility (GARCH / historical covariance) alongside sentiment tilts.
- Broker API Execution: Connect FIX protocol / Zerodha Kite for real-world systematic trade execution.